Conditioning the female athlete

KEEBRA PARK GRLA — WHAT CHANGES, AND WHAT DOES NOT

The principle The principles of training do not change for female athletes. The way they are applied must.

Do not assume a conditioning program built for male athletes transfers across.

FIVE DIFFERENCES THAT CHANGE THE SESSION

WHAT IS DIFFERENT	WHAT YOU DO ABOUT IT
Higher heart rate at the same relative work	Smaller heart and stroke volume means she beats faster for the same output. Do not judge her off a male HR benchmark.
Better at sustaining moderate work	More Type I fibres and about a third greater mitochondrial respiration. Repeat-effort work is a strength — use it.
Steeper power drop between maximal efforts	At maximal and supramaximal intensity she may fade faster between reps. Give slightly longer recovery, not shorter.
Luteal phase raises core temperature 0.3–0.5°C	Heat tolerance drops in the two weeks after ovulation. Move hard sessions to the cool part of the day.
Lactate threshold at a higher % of VO2max	She holds a higher percentage before she tips over. HR-based zones built for males will under-prescribe her.

WHAT THE GAME ASKS FOR women's rugby league, measured

MEASURE	NUMBER
Relative distance, NRLW match	About 79–80 m per minute. State of Origin runs higher at ~81.6, Trans-Tasman Tests lower at ~73.4.
Peak 30-second intensity	Backs ~189 m/min, forwards ~161 m/min. Over a 10-minute window both settle near 81–89.
A repeat high-intensity bout	Three or more efforts — sprints, tackles, rucks — inside 21 seconds. Longest bouts run 49–64 seconds.
When bouts happen	About 70% fall within five minutes of a try being scored. Condition for the critical periods, not the average.
Contact and running together	Running output drops as contact efforts rise. Training them separately trains neither.

THE ONE RULE THAT OUTRANKS THE SESSION PLAN

Technique rule: if running mechanics, landing or cutting start to fall apart, reduce the INTENSITY before you reduce the volume. Protective movement patterns fail earlier in female athletes than male at the same relative load (QRL Section 4). A missed bout costs one bout. A soft-tissue injury in week three costs the season.

Sources: QRL Female Athlete Physical Framework §4 · Gabbett 2005, 2006 · Austin, Gabbett & Jenkins 2011 · Johnston, Gabbett & Jenkins 2014 · NRLW GPS 2018–19